🎓 AI Viva Preparation Platform

Practice. Answer. Improve. Ace Your Viva.

🔗 **Live Application:** https://viva-prep-flow.base44.app

📌 Project Overview

AI Viva Preparation Platform is an AI-powered web application designed to help B.Tech and CSE students prepare for viva examinations.

The platform allows students to select a subject or provide their project details, generate personalized viva questions using AI, answer those questions, and receive performance evaluation, scores, and improvement feedback.

Instead of simply providing a list of questions, the platform aims to simulate an interactive viva experience where students can practice technical communication and identify their weak areas.

🎯 Problem Statement

Many students prepare for viva examinations by memorizing predefined questions and answers. This approach does not effectively test their understanding or ability to explain technical concepts.

Students need a platform where they can:

- Practice viva questions interactively
- Get questions according to their subject or project
- Answer questions without seeing the answer beforehand
- Receive immediate feedback
- Understand their weak areas
- Track their performance over multiple practice sessions

The **AI Viva Preparation Platform** addresses this problem by providing an AI-assisted personalized viva preparation environment.

💡 Proposed Solution

The system uses AI to create a personalized viva practice experience.

Basic Workflow

Student Login

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Select Subject / Project

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Choose Difficulty & Number of Questions

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AI Generates Viva Questions

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Student Answers Questions

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AI Evaluates Answers

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Score + Feedback

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Final Performance Report

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Performance History & Analytics

✨ Key Features

1. 🔐 Student Authentication

Students can create an account and securely access their personal viva preparation data.

2. 💻 Subject-Based Viva

Students can select subjects such as:

- Python
- Data Structures
- DBMS
- Operating Systems
- Computer Networks
- Artificial Intelligence
- Machine Learning
- Cybersecurity
- Web Development
- Software Engineering

3. 🥷 Project Viva

Students can enter details about their project, including:

- Project title
- Problem statement
- Technologies used
- Features
- Student contribution
- Project description

The system can use these details to generate project-specific viva questions.

4. 🤖 AI Question Generation

AI generates questions based on:

- Subject
- Project
- Difficulty level
- Number of questions
- Viva type
- Technical topics

5. ✍️ Answer Submission

Students answer questions through an interactive viva interface.

6. 🧠 AI Answer Evaluation

The student's answer can be evaluated based on factors such as:

- Technical correctness
- Relevance
- Completeness
- Conceptual understanding
- Clarity

7. 📊 Performance Score

Students receive an overall performance score after completing their viva session.

Example:

Overall Score: 84/100

Technical Knowledge: 88%

Concept Understanding: 82%

Answer Relevance: 86%

Completeness: 78%

8. 💬 Personalized Feedback

The platform provides:

- Strengths
- Weak areas
- Missing concepts
- Improvement suggestions
- Recommended topics for revision

9. 📊 Performance Analytics

Students can track:

- Previous viva scores
- Average score
- Best score
- Questions attempted
- Subject-wise performance
- Improvement over time

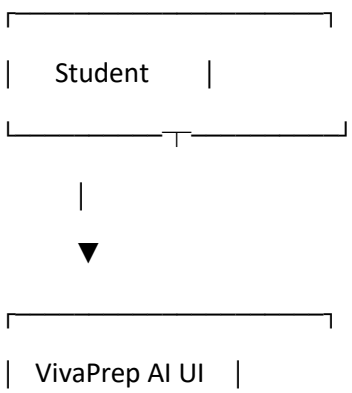
10. 🚀 Question Bookmarks

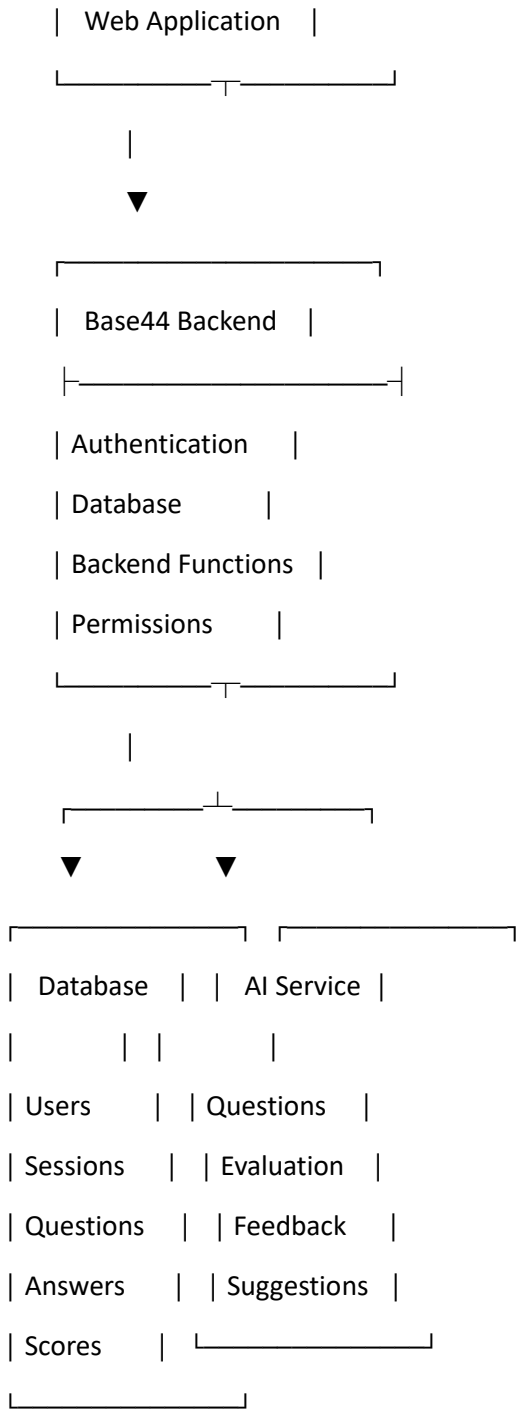
Students can save difficult questions and practice them later.

11. 🔄 Adaptive Viva Practice

The system can generate follow-up questions based on the student's previous answer to make the practice experience closer to a real viva.

🏗 System Architecture





Base44 provides built-in database, authentication, backend functions, integrations, permissions, and hosting capabilities for full-stack applications.

🛠 Technology Stack

Technology	Purpose
Base44 AI	Application development and AI-assisted development

React-based Web UI	Frontend interface
Base44 Database	Data storage
Base44 Authentication	User registration and login
Base44 Backend Functions	Server-side application logic
AI/LLM Integration	Question generation and answer evaluation
Base44 Hosting	Application deployment

Base44 provides an SDK for accessing entities, authentication, backend functions, and integrations from generated applications.

🗃 Main Data Entities

The application can maintain data for:

Users

Stores student account and profile information.

Subjects

Stores available academic subjects.

Projects

Stores student project information for project viva preparation.

Viva Sessions

Stores individual practice sessions.

Questions

Stores AI-generated or question-bank questions.

Answers

Stores student responses for a viva session.

Evaluations

Stores AI-generated evaluation and feedback.

Performance Reports

Stores final scores and performance metrics.

Bookmarks

Stores questions saved by students for future practice.

🔄 Application Workflow

Step 1 — Login

The student creates an account or logs into the platform.

Step 2 — Start Viva

The student selects:

- Subject or Project Viva
- Difficulty level
- Number of questions
- Question type

Step 3 — Question Generation

The AI generates relevant viva questions based on the selected information.

Step 4 — Answer

The student answers the question through the platform.

Step 5 — AI Evaluation

The answer is analyzed for technical understanding, correctness, relevance, and completeness.

Step 6 — Feedback

The student receives:

- Score
- Explanation
- Strengths
- Missing concepts
- Improvement suggestions

Step 7 — Final Report

After completing the session, the platform generates an overall performance report.

Step 8 — Track Progress

Previous sessions and scores can be reviewed through the history and analytics sections.

🎓 Educational Benefits

The application helps students:

- Improve technical knowledge
- Practice explaining concepts

- Develop confidence
- Identify weak topics
- Prepare for project viva
- Improve technical communication
- Practice answering unexpected questions
- Track their preparation progress

🚀 Future Enhancements

The following features can be added in future versions:

- 🎤 Voice-based viva
- 🗣️ Speech-to-text answers
- 🔊 AI examiner voice
- 😊 Confidence and communication analysis
- 📹 Video-based mock viva
- 👩‍🏫 Faculty-created question banks
- 🏆 Student leaderboard
- 📱 Mobile application
- 📄 AI-generated viva preparation notes
- 🌐 Multi-language viva support
- 🗂️ University-specific question banks
- 🔗 Integration with college LMS
- 📊 Advanced learning analytics

🔐 Security Considerations

The application should ensure:

- Secure authentication
- User-specific data access
- Role-based permissions
- Protected backend operations
- Secure handling of AI/API credentials

- Input validation
- Protection of student information

Base44 backend functions can run server-side and can access protected secrets without exposing them in the frontend.

🩹 Testing

The application should be tested for:

Functional Testing

- Login
- Registration
- Subject selection
- Viva creation
- Question generation
- Answer submission
- Evaluation
- Score calculation
- History

UI Testing

- Desktop
- Tablet
- Mobile

Validation Testing

- Empty answers
- Invalid inputs
- Invalid project information
- Session interruption

Performance Testing

- Question generation
- Answer evaluation
- Dashboard loading
- Analytics rendering

📸 Screenshots

Add screenshots of your application here.

Recommended screenshots:

1. Landing Page
2. Login Page
3. Student Dashboard
4. Start Viva Page
5. AI Generated Questions
6. Answer Evaluation
7. Final Performance Report
8. Viva History
9. Analytics Dashboard

Example:

📸 Screenshots

Dashboard

Viva Practice

Performance Report

🌐 Live Demo

VivaPrep AI:
https://viva-prep-flow.base44.app

🎓 Project Type

Mini Project — B.Tech Computer Science and Engineering

Domain

- Artificial Intelligence
- Web Development
- Educational Technology
- Natural Language Processing
- AI-Assisted Learning

🎯 Project Objective

The primary objective of the **AI Viva Preparation Platform** is to provide students with an interactive and personalized environment for practicing viva examinations using AI-generated questions and intelligent answer evaluation.

📌 Conclusion

The **AI Viva Preparation Platform** demonstrates how AI can be used to improve traditional examination preparation.

By combining an interactive web application with AI-based question generation, answer evaluation, personalized feedback, and performance analytics, the platform provides students with a more effective and engaging way to prepare for technical viva examinations.

The project also demonstrates how AI-assisted development platforms such as **Base44** can be used to create full-stack applications with database, authentication, backend logic, AI integrations, and deployment capabilities.

⭐ Project Highlights

Select → Practice → Answer → Evaluate → Improve

VivaPrep AI
Practice. Answer. Improve. Ace Your Viva.